

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 6289A	Course Title: Virtualization and Cloud Security Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Learn how to secure compute services viz. securing VMs, Managed Database Services and Securing server less/function as a service
- Practice securing cloud based IAM services, conducting security monitoring and audit trails
- Experiment with securing Storage Services.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
General knowledge about operating system commands	3157	System Administration Lab	3

Course Outcomes:

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate securing compute services	9	Applying
CO2	Experiment with securing networking services	9	Applying
CO3	Demonstrate securing cloud based IAM services	9	Applying
CO4	Experiment with securing storage services and conduct security monitoring and audit trails	15	Applying

	Lab Test	3	
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CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			3
CO2	3			3			3
CO3	3			3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate securing compute services		
M1.01	Secure Amazon Elastic Compute Cloud (EC2)	3	Applying
M1.02	Implement security in Amazon RDS for MySQL	3	Applying
M1.03	Demonstrate securing AWS Lambda	3	Applying
CO2	Experiment with securing networking services		
M2.01	Implement security in Amazon Virtual Private Cloud	6	Applying
M2.02	Secure Content Delivery Network Services	3	Applying
	Lab Test 1	1.5	
CO3	Demonstrate securing cloud-based IAM services		
M3.01	Demonstrate securing AWS IAM	3	Applying
M3.02	Experiment with auditing AWS IAM	3	Applying

M3.03	Secure AWS Directory Services	3	Applying
CO4	Experiment with securing storage services and Conduct security monitoring and audit trails		
M4.01	Secure Amazon Simple Storage Service	2	Applying
M4.02	Experiment with securing Amazon Elastic Block Store	3	Applying
M4.03	Conduct security monitoring and audit trails using AWS CloudWatch	2	Applying
M4.04	Perform security monitoring and audit trails using AWS CloudTrail	3	Applying
	Open ended experiments**	5	
	Lab Test – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

- Install Oracle virtual box and create two VMs on your Computer
- Install Turbo C in guest OS and execute C program
- Install Hadoop single node setup
- Create a guest book application.

Text / Reference

T/R	Book Title/Author
T1	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, Tim Mather, Subra Kumaraswamy, Shahed Latif, O'Reilly Media; 1st edition, 2009

T2	Cloud Security Handbook: Find out how to effectively secure cloud environments using AWS, Azure, and GCP, Eyal Estrin, Copyright © 2022 Packt Publishing
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Online Resources

Sl.No	Website Link
1	https://docs.aws.amazon.com/
2	https://docs.microsoft.com/en-us/cli/azure.
3	https://cloud.google.com/sdk